Tutorial - 2

(2) $\therefore$ ~~Let~~ $\left[\frac{1}{2}(A^{\dagger}+A)\right]^{\dagger} = \frac{(A^{\dagger})^{\dagger}+(A^{\dagger})^{\dagger}}{2}$

$\left[(A+B)^{\dagger} = A^{\dagger}+B^{\dagger}\right]$

$\frac{A^{\dagger}+A}{2}$ $\quad$ $\frac{A+A^{\dagger}}{2}$

$\therefore \left[\frac{1}{2}(A^{\dagger}-A^{\dagger})\right]^{\dagger}$ $\cancel{= \frac{(A^{\dagger})^{\dagger}-A^{\dagger}}{2} = \frac{A-A^{\dagger}}{2}}$

$= \frac{A^{\dagger}-(A^{\dagger})^{\dagger}}{2}$ $\quad$ $\frac{A^{\dagger}-A}{2}$

$\frac{A^{\dagger}+A}{2} \longrightarrow$ Hermitian

$\frac{A-A^{\dagger}}{2} \longrightarrow$ anti-Hermitian

$\therefore A = \frac{A+A^{\dagger}}{2} + \frac{A-A^{\dagger}}{2}$ = Hermitian + anti-Hermitian

① (a) $F = -\frac{dV}{dx}$.

$\Rightarrow \quad -\int F\,dx = \int dV.$

$\Rightarrow \quad \boxed{V(x) = -Fx + c}$

(b) TDSE: (position-space)

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi = i\hbar\frac{\partial\psi}{\partial t}$$

$$\Rightarrow \quad -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + (-Fx + c)\psi = i\hbar\frac{\partial\psi}{\partial t} \quad \text{— ①}$$

Take $\psi(x,t) = e^{-ict/\hbar}\,\phi(x,t)$.

$\therefore \frac{\partial\psi}{\partial x} = e^{-ict/\hbar}\frac{\partial\phi}{\partial x}$

$\therefore \frac{\partial^2\psi}{\partial x^2} = e^{-ict/\hbar}\frac{\partial^2\phi}{\partial x^2}$

$\therefore \frac{\partial\psi}{\partial t} = e^{-ict/\hbar}\left(\frac{\partial\phi}{\partial t} - \phi\frac{ic}{\hbar}\right)$.

Substituting into ①:

$$-\frac{\hbar^2}{2m}\cancel{e^{-ict/\hbar}}\frac{\partial^2\phi}{\partial x^2} + (-Fx + c)\cancel{e^{-ict/\hbar}}\phi(x,t) = i\hbar\cancel{e^{-ict/\hbar}}\left(\frac{\partial\phi}{\partial t} - \phi\frac{ic}{\hbar}\right)$$

$$\Rightarrow \quad -\frac{\hbar^2}{2m}\frac{\partial^2\phi}{\partial x^2} + (-Fx + \cancel{c})\phi = i\hbar\frac{\partial\phi}{\partial t} + \cancel{c\phi}$$

$$\Rightarrow \quad \boxed{-\left(\frac{\hbar^2}{2m}\frac{\partial^2\phi}{\partial x^2} + Fx\phi\right) = i\hbar\frac{\partial\phi}{\partial t}} \quad \text{— position-space representation} \quad \text{— ⑪}$$

This shows that the constant c is irrelevant (only induces a phase factor) & the true potential is $V(x) = -Fx$.

Momentum-space:

Applying $\langle p|$ to both sides of ⑪:

RHS: $i\hbar\langle p|\frac{\partial}{\partial t}|\phi\rangle = i\hbar\frac{\partial}{\partial t}\langle p|\phi\rangle = i\hbar\frac{\partial}{\partial t}\phi(p,t)$

$\langle p|\frac{\partial}{\partial t}|\phi\rangle = \frac{\partial}{\partial t}\langle p|\phi\rangle$ is allowed because both the operator $(\frac{\partial}{\partial t})$ & the bra $\langle p|$ are independent of time

In general:

$$\frac{\partial}{\partial t}\langle A|B\rangle = \langle \dot{A}|B\rangle + \langle A|\dot{B}\rangle.$$

$\because \dot{A} = 0$; $\frac{\partial}{\partial t}\langle A|B\rangle = \langle p|\frac{\partial}{\partial t}|\phi\rangle$

LHS: (i) Kinetic Energy Term:

$$\langle p|-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}|\phi\rangle = -\frac{\hbar^2}{2m}\langle p|\frac{\partial^2}{\partial x^2}|\phi\rangle$$

$$\cancel{= \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\phi(p,t)}$$

$$= -\frac{\hbar^2}{2m}\frac{1}{\sqrt{2\pi\hbar}}\underbrace{\int_{-\infty}^{\infty}\frac{\partial^2}{\partial x^2}\psi(x,t)\, e^{-ipx/\hbar}\,dx}_{I_1}$$

$$\therefore I_1 = \left[e^{-ipx/\hbar}\frac{\partial\psi(x,t)}{\partial x}\right]_{-\infty}^{\infty} + \frac{ip}{\hbar}\int e^{-ipx/\hbar}\frac{\partial\psi(x,t)}{\partial x}\,dx.$$

$$= \frac{ip}{\hbar}\left[\left[e^{-ipx/\hbar}\psi(x,t)\right]_{-\infty}^{\infty} + \frac{ip}{\hbar}\int_{-\infty}^{\infty} e^{-ipx/\hbar}\psi(x,t)\right]$$

$$= -\frac{p^2}{\hbar^2}\int_{-\infty}^{\infty} e^{-ipx/\hbar}\psi(x,t)\,dx.$$

$$\therefore \frac{\cancel{-}\hbar^2}{2m}\cdot\frac{1}{\sqrt{2\pi\hbar}}\cdot\frac{\cancel{-}p^2}{\cancel{\hbar^2}}\int_{-\infty}^{\infty} e^{-ipx/\hbar}\psi(x,t)\,dx.$$

$$= \boxed{\frac{p^2}{2m}\phi(p,t)}$$

(ii) Potential Term:

$$\langle p|-Fx|\phi\rangle = -F\langle p|x|\phi\rangle.$$

$$\cancel{= Fx\langle p|\phi\rangle = Fx\,\phi(p,t)}$$

$$= -F\frac{1}{\sqrt{2\pi\hbar}}\underbrace{\int_{-\infty}^{\infty} e^{-ipx/\hbar}\, x\,\psi(x,t)\,dx}_{I_2}$$

$\cancel{\therefore I_2 =}$

Now, $\phi(p,t) = \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty} e^{-ipx/\hbar}\psi(x,t)\,dx.$

$$\Rightarrow \frac{\partial\phi}{\partial p} = \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}\psi(x,t)\frac{\partial}{\partial p}\left(e^{-ipx/\hbar}\right)dx.$$

$$= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \psi(x,t) \left(-\frac{ix}{\hbar}\right) e^{-ipx/\hbar}\, dx.$$

$$= -\frac{i}{\hbar} \frac{1}{\sqrt{2\pi\hbar}} \boxed{\left(\int_{-\infty}^{\infty} \psi(x,t) \times e^{-ipx/\hbar}\, dx\right)} \rightarrow I_2.$$

$$\therefore \quad \frac{\partial \phi}{\partial p} = -\frac{i}{\hbar} \frac{1}{\sqrt{2\pi\hbar}} \cdot I_2$$

$$\Rightarrow \quad I_2 = i\hbar \sqrt{2\pi\hbar} \cdot \frac{\partial \phi}{\partial p}$$

$$\therefore \quad \langle p | Fx | \phi \rangle = -F \cdot \frac{1}{\sqrt{2\pi\hbar}} \cdot i\hbar \sqrt{2\pi\hbar} \cdot \frac{\partial \phi}{\partial p}$$

$$= \boxed{-i\hbar F \frac{\partial \phi(p,t)}{\partial p}}$$

Combining both the terms

$$i\hbar \frac{\partial \phi(p,t)}{\partial t} = \frac{p^2}{2m} \phi(p,t) - i\hbar F \frac{\partial \phi(p,t)}{\partial p}$$

$$\Rightarrow \quad \boxed{\frac{\partial \phi(p,t)}{\partial t} = -\frac{i}{\hbar} \frac{p^2}{2m} \phi(p,t) - F \frac{\partial \phi(p,t)}{\partial p}} \quad \text{—— (iii)}$$

↳ momentum-space representat^n.

(c) Let $\phi(p,t)$ be a stationary state (energy eigenstate) of the form:

$$\phi(p,t) = \psi(p)\, e^{-iEt/\hbar}$$

$$\therefore \quad \frac{\partial \phi(p,t)}{\partial t} = \psi(p) \left(-\frac{iE}{\hbar}\right) e^{-iEt/\hbar}$$

$$\therefore \quad \frac{\partial \phi}{\partial p}(p,t) = e^{-iEt/\hbar} \frac{\partial \psi(p)}{\partial p}$$

Substituting in (iii):

$$\psi(p)\, e \left(+\frac{iE}{\hbar}\right) \cancel{e^{-iEt/\hbar}} = +\frac{i}{\hbar} \frac{p^2}{2m} \psi(p) \cancel{e^{-iEt/\hbar}} + F \cancel{e^{-iEt/\hbar}} \frac{\partial \psi(p)}{\partial p}.$$

$$\Rightarrow \quad \cancel{\psi(p)} \boxed{\frac{\partial \psi}{\partial p} + \left[\frac{ip^2}{\hbar 2mF} - \frac{iE}{F\hbar}\right] \psi(p) = 0}$$

$\therefore$ ~~$\psi(p) = Ne^{iAp}$~~

$$\psi_F = e^{\int (ip^2/2mF\hbar - iE/\hbar F)\, dp}$$

$$= \exp\left[\frac{i}{\hbar F}\left(\frac{p^3}{6m} - Ep\right)\right]$$

$$\therefore \quad \psi \exp\left[\frac{i}{\hbar F}\left(\frac{p^3}{6m} - Ep\right)\right]$$

$$\Rightarrow \quad \boxed{\psi_E(p) = N e^{\frac{i}{\hbar F}\left(Ep - \frac{p^3}{6m}\right)}} \quad \text{(Ans)}$$

(d) $$\int \psi_{E'}^{*}(p)\, \psi_E(p)\, dp. = I$$

$$= |N|^2 \int \exp\left[-\frac{i}{\hbar F}\left(E'p - \frac{p^3}{6m} - Ep + \frac{p^3}{6m}\right)\right] dp.$$

$$= |N|^2 \int_{-\infty}^{\infty} e^{\frac{i}{\hbar F}\left(\frac{E-E'}{\hbar F}\right)p}\, dp.$$

According to the identity ; $x = \dfrac{E - E'}{\hbar F}$.

$$\therefore \quad I = 2\pi |N|^2\, \delta\left(\frac{E-E'}{\hbar F}\right)$$

$$= 2\pi |N|^2 \hbar |F|\, \delta(E-E')$$

Now, $$I = \delta(E-E')$$

$$\therefore \quad 2\pi |N|^2 \hbar |F| = 1$$

$$\Rightarrow \quad |N| = \frac{1}{\sqrt{2\pi\hbar|F|}}$$

An overall ~~phas~~ arbitrary phase of N makes :

$$\boxed{N = \frac{1}{\sqrt{2\pi\hbar|F|}}\, e^{i\theta}}$$